

Firm nonexpansiveness and fixed points of resolvents

Proof of Conjecture 00000008844

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Definitions and domain

Let H be a real Hilbert space and let $A : H \rightrightarrows H$ be monotone: for every $a \in A(u)$ and $b \in A(v)$,

$$\langle u - v, a - b \rangle \geq 0.$$

For the standard positive parameter $\lambda > 0$, define the resolvent $J_\lambda = (I + \lambda A)^{-1}$ by

$$u \in J_\lambda(x) \iff \exists a \in A(u) : x = u + \lambda a.$$

Its natural domain is $D_\lambda = \text{ran}(I + \lambda A)$. We will prove that its values there are singletons, so it is an actual map $J_\lambda : D_\lambda \rightarrow H$. The source conjecture's two properties hold on this natural domain. No claim of totality on H is made: monotonicity alone does not give that additional property. For example, the graph containing only $(0, 0)$ is monotone and its resolvent domain is $\{0\}$.

Firm nonexpansiveness

Take $x, y \in D_\lambda$, $u \in J_\lambda(x)$, and $v \in J_\lambda(y)$. Choose $a \in A(u)$ and $b \in A(v)$ with $x = u + \lambda a$, $y = v + \lambda b$. Then

$$\begin{aligned} \langle u - v, x - y \rangle &= \langle u - v, (u - v) + \lambda(a - b) \rangle \\ &= \|u - v\|^2 + \lambda \langle u - v, a - b \rangle \geq \|u - v\|^2. \end{aligned}$$

If $x = y$, the inequality forces $\|u - v\|^2 = 0$, hence $u = v$. Existence on D_λ follows from its definition, and thus the resolvent is single-valued there. Substituting its unique values yields

$$\|J_\lambda(x) - J_\lambda(y)\|^2 \leq \langle J_\lambda(x) - J_\lambda(y), x - y \rangle,$$

which is the defining firm nonexpansiveness inequality.

Fixed points and zeros

For any $x \in H$, the diagonal graph relation is equivalent to a zero:

$$x \in J_\lambda(x) \iff \exists a \in A(x) : x = x + \lambda a \iff 0 \in A(x).$$

The second equivalence uses $\lambda > 0$, hence $\lambda a = 0$ implies $a = 0$. A zero therefore automatically belongs to D_λ . By single-valuedness, the displayed equivalence proves the exact set identity

$$\{x \in D_\lambda : J_\lambda(x) = x\} = \{x \in H : 0 \in A(x)\}.$$

Formal verification

Lean defines the set-valued graph, monotonicity, inverse graph, natural domain, and a choice-defined resolvent map. It proves the map satisfies the inverse graph and that this graph is exactly its value relation. The theorems `resolvent_firm` and `fixed_set_eq_zero_set` establish both original clauses. All results are for arbitrary real inner-product spaces, so they include Hilbert spaces without needing completeness or maximality.